# UTKARSH SINGH

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## Professional Summary

Undergrad AI researcher specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning. I care about taste in engineering. From hackathon floors to research labs, I ship fast, break hard problems into working prototypes, and care deeply about systems that solve real user pain.

## Education

**Maharaja Agrasen Institute of Technology [GGSIPU]** — New Delhi, India
B.Tech in Computer Science Engineering, Specialization in AI (9.14 CGPA) — Aug. 2025 - May 2029

**Kamal Public Sr. Sec. School [CBSE]** — New Delhi, India
Class XII PCM with Computer Science (92.4%) — April 2011 - May 2025

## Experience

**AI and Software Engineer Intern**, The Veritas (Remote) — Oct 2026 - Present

- Shipping AI features and backend services across the product; currently building a real time market data platform (BTC/USD, NSE India, news flow) for investor demos.
- Working across the full stack: backend APIs, databases, frontend and the AI models powering investor insights.

**Research Intern**, Maharaja Agrasen Institute of Technology (Remote) — Jun 2026 - Present

- Working on multiple papers in parallel, from LLM inference systems to trustworthy medical AI.
- Researching learned cache replacement for memory bound LLM decoding on CPUs, across dense and MoE models.
- Building VERA, a hallucination detector for radiology reports that checks attention against claimed anatomy.

## Research Publications

**Adaptive Termination in Agentic Multimodal RAG via Cross Modal Evidence Grounding**
[Accepted at ICML-SCALE 2026]

- Proposed CMEGS, a claim level cross modal grounding metric, and MIR, a provably terminating stopping rule.
- 90% faithfulness on PathVQA with 71% less hallucination than vanilla RAG; 2.3x fewer iterations on ScienceQA.

## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, C++, C
**AI and ML:** Generative AI, Agentic AI, Multi Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, PyTorch, TensorFlow, Transformers, Hugging Face, Scikit Learn
**Backend and APIs:** FastAPI, Flask, Node.js, Express.js, REST APIs
**Frontend:** React, Next.js, Tailwind CSS
**Databases:** PostgreSQL, MySQL, MongoDB, Redis, Firebase, SQLite, DynamoDB
**MLOps and Cloud:** Docker, GitHub Actions, AWS (S3, EC2, Lambda, Bedrock), GCP, Vercel, Render, Railway
**Libraries and Tools:** NumPy, Pandas, LangChain, Matplotlib, OpenCV, ONNX Runtime, Postman, Jupyter

## Projects

**KAVACH: Fraud Risk Manager for Indian D2C Merchants**

- Python, FastAPI, XGBoost, SHAP, NetworkX, Next.js: built a calibrated return risk scorer paired with graph based fraud ring detection over shared phones, devices, VPAs and fuzzy matched addresses.
- Defence only risk manager for Indian D2C merchants targeting COD return abuse and coordinated chargebacks that erode 5 to 15% of gross margin.
- Ring detection scored 0.889 precision at 1.000 recall on held out data; every flagged order ships with SHAP attributions and a SHA256 hash chained audit trail.

**DelhiJalNav: Flood Aware Route Planner for Delhi**

- TypeScript, Python, AWS (Lambda, DynamoDB, API Gateway, CloudFront, Bedrock), ONNX Runtime Web: built a serverless PWA that checks candidate routes against crowd reported waterlogging, with Bedrock drafting reports from Hindi, English or Hinglish text.
- Steers Delhi commuters around flooded underpasses using an evidence policy where each hotspot moves through reported, corroborated, conflicted and stale states, so only a reviewed clear can unblock a route.
- Runs YOLOv8n pothole detection and SegFormer water segmentation on device in the browser, with an offline safe IndexedDB outbox; 58/58 backend tests and 12/12 live smoke tests pass.

**NETRA: AI Growth Copilot for Indian Kirana Stores**

- Python, FastAPI, Llama 3.3 70B, Sarvam AI, React, Differential Privacy: built demand forecasting, non collusive price benchmarking and credit scoring delivered via Soundbox audio, WhatsApp and web in 7 Indic languages.
- AI growth copilot for Indian Kirana stores serving 90% of a $600B+ grocery market, enabling solo merchants to make data backed pricing and inventory decisions.
- Enforces a five layer privacy boundary using k anonymity cohort gating and Laplace DP to block competitor surveillance and prevent price collusion.